

Further Applications of Integration (Chapter 8)

8.1 Arc Length

The Arc Length Formula If f' is continuous on $[a, b]$, then the length of the curve $y = f(x)$, $a \leq x \leq b$, is

$$L = \int_a^b \sqrt{1 + [f'(x)]^2} \, dx$$

If we use Leibniz notation for derivatives, we can write the arc length formula as follows:

$$L = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2} \, dx$$

If a curve has the equation $x = g(y)$, $c \leq y \leq d$, and $g'(y)$ is continuous, then by interchanging the roles of x and y we obtain the following formula for its length:

$$L = \int_c^d \sqrt{1 + [g'(y)]^2} \, dy = \int_c^d \sqrt{1 + \left(\frac{dx}{dy}\right)^2} \, dy$$

The Arc Length Function

We will find it useful to have a function that measures the arc length of a curve from a particular starting point to any other point on the curve. Thus if a smooth curve C has the equation $y = f(x)$, $a \leq x \leq b$, let $s(x)$ be the distance along C from the initial point $P_0(a, f(a))$ to the point $Q(x, f(x))$. Then s is a function, called the **arc length function**,

$$s(x) = \int_a^x \sqrt{1 + [f'(t)]^2} \, dt$$

(We have replaced the variable of integration by t so that x does not have two meanings.) We can use Part 1 of the Fundamental Theorem of Calculus to differentiate Equation 5 (since the integrand is continuous):

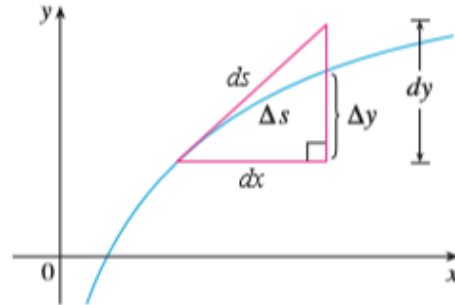
$$\frac{ds}{dx} = \sqrt{1 + [f'(x)]^2} = \sqrt{1 + \left(\frac{dy}{dx}\right)^2}$$

Equation shows that the rate of change of s with respect to x is always at least 1 and is equal to 1 when $f'(x)$, the slope of the curve, is 0. The differential of arc length is

$$ds = \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

and this equation is sometimes written in the symmetric form

$$(ds)^2 = (dx)^2 + (dy)^2$$



The geometric interpretation is shown in Figure

If we write $L = \int ds$, then we can solve to get

$$ds = \sqrt{1 + \left(\frac{dx}{dy}\right)^2} dy$$

8.2 Area of a Surface of Revolution

in the case where f is positive and has a continuous derivative, we define the **surface area** of the surface obtained by rotating the curve $y = f(x)$, $a \leq x \leq b$, about the x -axis as

$$S = \int_a^b 2\pi f(x) \sqrt{1 + [f'(x)]^2} dx$$

With the Leibniz notation for derivatives, this formula becomes

$$S = \int_a^b 2\pi y \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

If the curve is described as $x = g(y)$, $c \leq y \leq d$, then the formula for surface area becomes

$$S = \int_c^d 2\pi y \sqrt{1 + \left(\frac{dx}{dy}\right)^2} dy$$

and both Formulas can be summarized symbolically, using the notation for arc length given in Section 8.1, as

$$S = \int 2\pi y \, ds$$

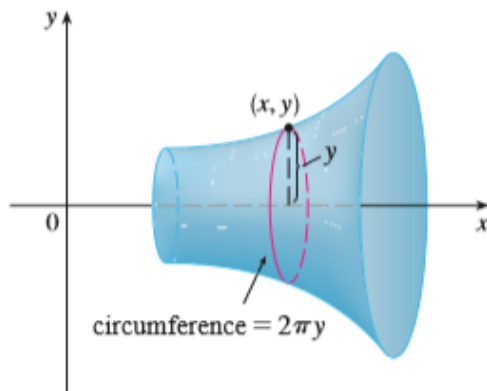
For rotation about the y-axis, the surface area formula becomes

$$S = \int 2\pi x \, ds$$

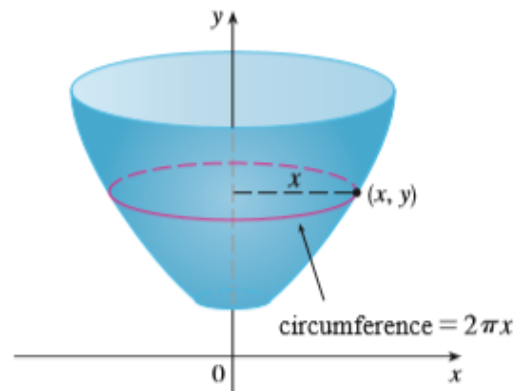
where, as before, we can use either

$$ds = \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx \quad \text{or} \quad ds = \sqrt{1 + \left(\frac{dx}{dy}\right)^2} dy$$

These formulas can be remembered by thinking of $2\pi y$ or $2\pi x$ as the circumference of a circle traced out by the point (x, y) on the curve as it is rotated about the x -axis or y -axis, respectively



(a) Rotation about x -axis: $S = \int 2\pi y \, ds$



(b) Rotation about y -axis: $S = \int 2\pi x \, ds$

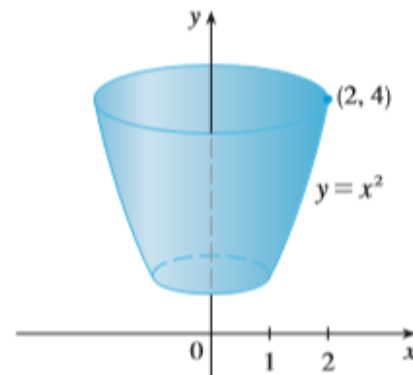
EXAMPLE The arc of the parabola $y = x^2$ from $(1, 1)$ to $(2, 4)$ is rotated about the y -axis. Find the area of the resulting surface.

SOLUTION Using

$$y = x^2 \quad \text{and} \quad \frac{dy}{dx} = 2x$$

we have

$$\begin{aligned} S &= \int 2\pi x \, ds = \int_1^2 2\pi x \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx \\ &= \dots\dots\dots \end{aligned}$$



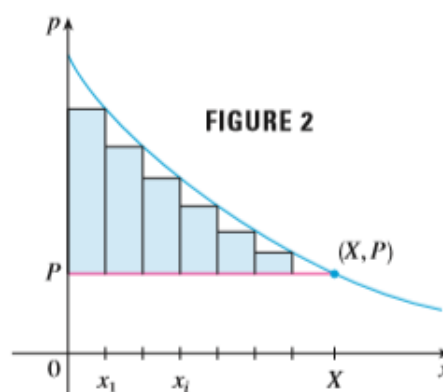
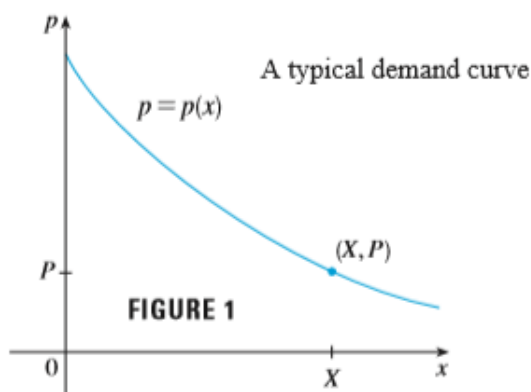
8.3 Applications to Physics and Engineering

.... (Please skip sections 8.3 and 8.4!)

8.4 Applications to Economics and Biology

Consumer Surplus

Recall from Section 4.7 that the demand function $p(x)$ is the price that a company has to charge in order to sell x units of a commodity. Usually, selling larger quantities requires lowering prices, so the demand function is a decreasing function. The graph of a typical demand function, called a **demand curve**, is shown in Figure 1. If X is the amount of the commodity that is currently available, then $P = p(X)$ is the current selling price.



We divide the interval $[0, X]$ into n subintervals, each of length $\Delta x = X/n$, and let $x_i^* = x_i$ be the right endpoint of the i th subinterval, as in Figure 2. If, after the first x_{i-1} units were sold, a total of only x_i units had been available and the price per unit had been set at $p(x_i)$ dollars, then the additional Δx units could have been sold (but no more). The consumers who would have paid $p(x_i)$ dollars placed a high value on the product; they would have paid what it was worth to them. So in paying only P dollars they have saved an amount of

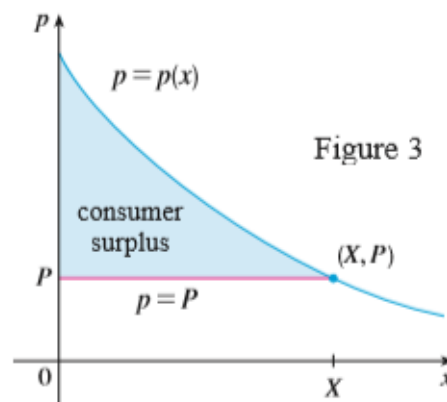
$$(\text{savings per unit})(\text{number of units}) = [p(x_i) - P] \Delta x$$

Considering similar groups of willing consumers for each of the subintervals and adding the savings, we get the total savings:

$$\sum_{i=1}^n [p(x_i) - P] \Delta x$$

(This sum corresponds to the area enclosed by the rectangles in Figure 2.) If we let $n \rightarrow \infty$, this Riemann sum approaches the integral

$$\int_0^X [p(x) - P] dx$$



which economists call the **consumer surplus** for the commodity.

The consumer surplus represents the amount of money saved by consumers in purchasing the commodity at price P , corresponding to an amount demanded of X . Figure 3 shows the interpretation of the consumer surplus as the area under the demand curve and above the line $p = P$.

(Read next one!)

8.5 Probability

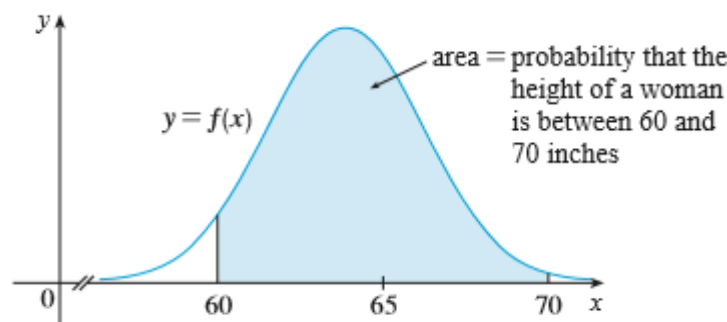
Every continuous random variable X has a **probability density function** f . This means that the probability that X lies between a and b is found by integrating f from a to b :

$$P(a \leq X \leq b) = \int_a^b f(x) dx$$

In general, the probability density function f of a random variable X satisfies the condition $f(x) \geq 0$ for all x . Because probabilities are measured on a scale from 0 to 1, it follows that

$$\int_{-\infty}^{\infty} f(x) dx = 1$$

Probability density function
for the height of an adult female



EXAMPLE Let $f(x) = 0.006x(10 - x)$ for $0 \leq x \leq 10$ and $f(x) = 0$ for all other values of x .

- Verify that f is a probability density function.
- Find $P(4 \leq X \leq 8)$.

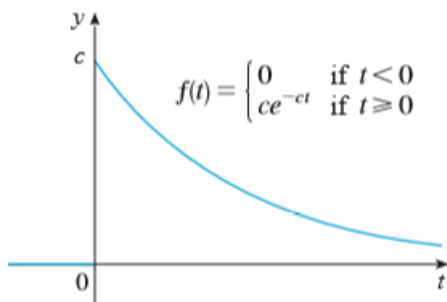
EXAMPLE Phenomena such as waiting times and equipment failure times are commonly modeled by exponentially decreasing probability density functions. Find the exact form of such a function.

SOLUTION Think of the random variable as being the time you wait on hold before an agent of a company you're telephoning answers your call. So instead of x , let's use t to represent time, in minutes. If f is the probability density function and you call at time $t = 0$, then, from Definition 1, $\int_0^2 f(t) dt$ represents the probability that an agent answers within the first two minutes and $\int_4^5 f(t) dt$ is the probability that your call is answered during the fifth minute.

It's clear that $f(t) = 0$ for $t < 0$ (the agent can't answer before you place the call). For $t > 0$ we are told to use an exponentially decreasing function, that is, a function of the form $f(t) = Ae^{-ct}$, where A and c are positive constants. Thus

$$f(t) = \begin{cases} 0 & \text{if } t < 0 \\ Ae^{-ct} & \text{if } t \geq 0 \end{cases}$$

We determine the value of A :



An exponential density function

every exponential density function has the form

$$f(t) = \begin{cases} 0 & \text{if } t < 0 \\ ce^{-ct} & \text{if } t \geq 0 \end{cases}$$

Average Values

In general, the **mean** of any probability density function f is defined to be

$$\mu = \int_{-\infty}^{\infty} x f(x) dx$$

The mean can be interpreted as the long-run average value of the random variable X . It can also be interpreted as a measure of centrality of the probability density function.

The mean is $\mu = 1/c$, so we can rewrite the probability density function as

$$f(t) = \begin{cases} 0 & \text{if } t < 0 \\ \mu^{-1} e^{-t/\mu} & \text{if } t \geq 0 \end{cases}$$

EXAMPLE Suppose the average waiting time for a customer's call to be answered by a company representative is five minutes.

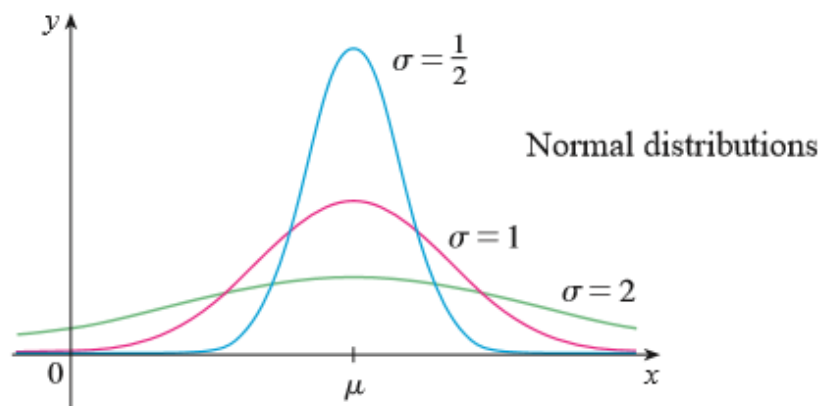
- (a) Find the probability that a call is answered during the first minute.
- (b) Find the probability that a customer waits more than five minutes to be answered.

Normal Distributions

Many important random phenomena—such as test scores on aptitude tests, heights and weights of individuals from a homogeneous population, annual rainfall in a given location—are modeled by a **normal distribution**. This means that the probability density function of the random variable X is a member of the family of functions

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-(x-\mu)^2/(2\sigma^2)}$$

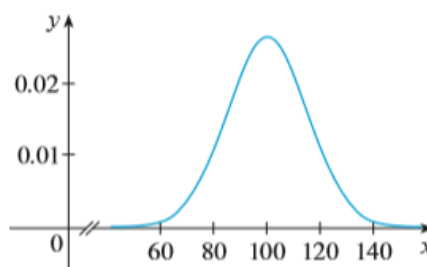
You can verify that the mean for this function is μ . The positive constant σ is called the **standard deviation**; it measures how spread out the values of X are. From the bell-shaped graphs of members of the family in Figure, we see that for small values of σ the values of X are clustered about the mean, whereas for larger values of σ the values of X are more spread out. Statisticians have methods for using sets of data to estimate μ and σ .



The factor $1/(\sigma\sqrt{2\pi})$ is needed to make f a probability density function. In fact, it can be verified using the methods of multivariable calculus that

$$\int_{-\infty}^{\infty} \frac{1}{\sigma\sqrt{2\pi}} e^{-(x-\mu)^2/(2\sigma^2)} dx = 1$$

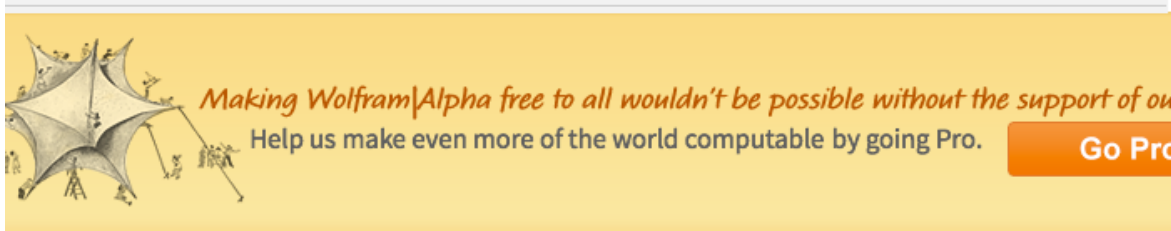
EXAMPLE Intelligence Quotient (IQ) scores are distributed normally with mean 100 and standard deviation 15.



Distribution of IQ scores

- What percentage of the population has an IQ score between 85 and 115?
- What percentage of the population has an IQ above 140?

wolframalpha.com/input/?i=integrate+1/(15*sqrt(2*pi))*exp(-(x-100)^2/450)+dx+from+x=140+to+200



integrate 1/(15*sqrt(2*pi)) * exp(-(x-100)^2/450) dx from x=140 to 200



Web Apps Examples Random

Definite integral:

More digits

$$\int_{140}^{200} \frac{\exp\left(-\frac{1}{450}(x-100)^2\right)}{15\sqrt{2\pi}} dx = \frac{1}{2} \left(\operatorname{erf}\left(\frac{10\sqrt{2}}{3}\right) - \operatorname{erf}\left(\frac{4\sqrt{2}}{3}\right) \right) \approx 0.00383038$$

$\operatorname{erf}(x)$ is the error function

Therefore $0.0038 \approx$ about 0.4% of the population has an IQ over 140.